Credit Card Fraud Detection & Anomaly Audit Report

Production Intelligence Report | Generated: October 01, 2026 - 05:34:04 UTC | Strict Compliance

1. Executive Transaction Summary

Total Scored	**602**	**Flagged / Anomalous**	**39**
Anomaly Rate	**2.82%**	**Avg Transaction**	**$106.82**
Active Model	**Isolation Forest**	**Model Version**	**v1.0.0-test**

2. Active Model Performance & Benchmark Metrics

Metric	Score	Benchmark Guideline	Operational Status
Precision	0.8800	Minimizes false customer card declines	**Optimal**
Recall (Sensitivity)	0.8500	Maximizes fraud dollar capture	**Optimal**
F1-Score	0.9000	Harmonic balance of Precision & Recall	**Robust**
PR-AUC (Avg Precision)	0.9500	Primary metric for extreme imbalance	**High**
ROC-AUC	0.9400	Overall ranking capability across thresholds	**Superior**
False Positive Rate (FPR)	0.0150	Under 2% friction on genuine cardholders	**Within SLA**

3. Continuous Learning & Data Drift Diagnostics

Diagnostic Check	Observed Value	Threshold SLA	Automated Action
Population Stability (PSI)	5.0680	< 0.25 (Standard Threshold)	**RETRAINING_TRIGGERED**
Retraining Mode	Batch & Incremental Trigger	Continuous Drift Sentinel	**Enabled**
Last Retraining Event	Baseline Deployment	Scheduled / Drift Triggered	**SUCCESS**

4. Sample Flagged Transactions & Root-Cause Explainability

TX ID	Amount	Merchant / Channel	Score	Classification	Contributing Risk Factors
TX-1790813044.846638	$93.02	restaurant (Card)	**0.076**	**Normal**	[]
TX-1790813023.490214	$93.02	restaurant (Card)	**0.018**	**Normal**	[]
TX-1790812965.416952	$66.74	restaurant (Card)	**0.079**	**Normal**	[]
TX-1790812965.416902	$36.86	supermarket (Card)	**0.220**	**Normal**	[]
TX-1790812965.416852	$24.09	restaurant (Card)	**0.151**	**Normal**	[]

TX-1790812965.416803	$83.90	fuel_gas (Card)	**0.203**	Normal	[]
TX-1790812965.416753	$64.47	supermarket (Card)	**0.231**	Normal	[]
TX-1790812965.4167	$195.85	supermarket (Card)	**0.107**	Normal	[]
TX-1790812965.416638	$16.36	entertainment (Card)	**0.220**	Normal	[]
TX-1790812965.416587	$13.30	pharmacy (Card)	**0.357**	Normal	[]